SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

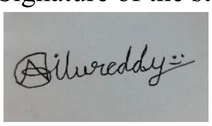
Dave's Taxonomy: Articulation (Level 4) **Total Marks: 30**
Question Count: 10

Code of Conduct

I G. Venkata Niranjan Reddy (192525199) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.
10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

ANSWERS

1. Unified Big Data Architecture for Retail

A multinational retail organization can integrate data from online websites, mobile apps, POS systems, loyalty programs, and social media into a centralized Big Data Lake. Tools such as Kafka can collect real-time data, while Spark can process and analyze customer behaviour. Machine-learning models can use purchase history and browsing patterns to provide personalized product recommendations and offers. This eliminates fragmented data and enables faster, data-driven decisions.

Data from all channels can first be collected and integrated into a common platform. Apache Kafka can handle real-time events such as purchases, product searches, and website clicks. Batch data from existing databases can be transferred using ETL tools.

A cloud data lake can store structured and unstructured customer information. Apache Spark can process large volumes of historical and current data. Customer profiles can be created by combining online and offline activities.

Example: If a customer searches for a laptop online and previously purchased computer accessories in a physical store, the system can recommend suitable laptops in real time.

2. Distributed Computing for Scientific Research

Distributed computing divides a large scientific dataset or computational task among multiple connected computers. Frameworks such as Hadoop and Apache Spark enable parallel processing, where several tasks are executed simultaneously. This significantly reduces processing time compared with a centralized system and allows resources to be increased when the dataset grows. It also improves scalability and fault tolerance.

The main advantages are:

- Faster data processing
- Ability to handle very large datasets
- Better scalability
- Reduced workload on a single system
- Improved fault tolerance

A centralized system is easier to manage for small datasets, but it becomes inefficient when data size increases significantly. Distributed computing is more suitable for large scientific projects because additional computing nodes can be added when required.

Example: Processing millions of satellite images can be divided among hundreds of computers instead of processing them sequentially on one machine.

3. Big Data Fraud Detection Framework

A digital payment platform can collect transaction data continuously using Apache Kafka and process it using Spark Streaming. Machine-learning models can analyze features such as transaction amount, location, device, frequency, and unusual spending patterns to identify fraudulent transactions. Real-time processing enables immediate alerts or transaction blocking, improving fraud detection accuracy and response speed.

Streaming processing is useful because the transaction can be analyzed immediately instead of waiting for a batch process. Predictive models improve detection by learning from previous fraud cases and customer behavior.

This approach increases fraud detection accuracy and allows suspicious transactions to be blocked or verified quickly.

Example: If a customer's card is used in Chennai and a few minutes later for a high-value transaction in another country, the system can identify the unusual pattern and flag it as suspicious.

4. Multimedia Data Management

Multimedia data such as videos, images, and audio creates challenges because of its huge volume, different file formats, and high processing requirements. Distributed storage such as HDFS or cloud object storage can store large multimedia files efficiently. Spark and other distributed processing technologies can perform parallel analysis, while compression reduces storage requirements.

The main challenges include:

- Large file sizes
- High storage requirements
- Different file formats
- Slow data transfer

Object storage is generally suitable for storing large multimedia files, while databases can store information such as file name, location, timestamp, and user details.This combination provides scalable storage and efficient processing of large multimedia datasets.

Example: A video-streaming company can store millions of videos in cloud object storage and use distributed processing to analyze viewing patterns.

5. Disaster Recovery and Business Continuity

A cloud-based Big Data platform should use data replication, backups, redundancy, and geographically distributed systems. Multiple copies of important data can be maintained in different locations so that data remains available even if one server or data center fails. Automated failover can redirect users and applications to a backup system without significant interruption.

The main components of the strategy include:

- Data replication
- Multiple availability zones
- Regular backups
- Disaster recovery sites
- Automated failover

Example: If the primary cloud data centre fails, the system automatically switches to a replicated secondary data center, reducing downtime and preventing major data loss.

6. Data Quality and Governance

Poor data quality, such as duplicate, inconsistent, or missing values, can produce incorrect analytics and poor business decisions. Data governance should include data validation, standardization, deduplication, quality monitoring, access control, and defined data ownership. These practices ensure that data remains accurate, consistent, secure, and trustworthy.

Duplicate records can be identified using unique IDs and matching techniques. Missing values can be corrected, estimated, or marked for review depending on their importance.

Master Data Management can maintain consistent information about important entities such as customers, products, and suppliers.

Example: If the same customer is stored twice with different addresses, deduplication can combine the records and prevent incorrect customer analysis.

7. Scalable Recommendation System

An online streaming service can collect watch history, ratings, searches, likes, and viewing duration from millions of users. Distributed technologies such as Kafka, Hadoop, and Spark can process this large volume of behavioral data. Machine-learning recommendation algorithms can identify user preferences and recommend relevant content, improving personalization, user engagement, and retention.

Apache Spark can process large numbers of user interactions and identify patterns.

Recommendation techniques can include:

- Collaborative Filtering
- Content-Based Filtering
- Matrix Factorization
- Deep Learning Models

Collaborative filtering recommends content based on users with similar interests. Content-based filtering recommends items similar to content the user has already watched.

Distributed processing allows millions of users and items to be analyzed efficiently. Real-time user behavior can also update recommendations.

Example: If a user frequently watches action movies, the system can analyze similar users' behavior and recommend new action movies or series.

8. Role of Cloud Computing in Big Data

Cloud computing provides on-demand storage, processing power, and scalability, making it highly suitable for Big Data applications. Organizations can increase or decrease resources according to workload without purchasing large amounts of hardware. It also provides flexibility and supports services such as cloud databases and analytics platforms. However, challenges include security, privacy, vendor dependency, and uncontrolled costs.

Advantages

Scalability: Resources can be increased or decreased based on demand.

Flexibility: Different services can be selected according to application requirements.

Reduced Initial Cost: Organizations do not need to purchase and maintain all servers.

Global Access: Cloud systems can provide services from multiple geographic locations.

Managed Services: Cloud providers offer managed databases, storage, analytics, and machine learning services.

Example: An e-commerce company can temporarily increase cloud computing resources during a festival sale when customer traffic and data processing requirements are high.

9. Real-Time Transportation Analytics

A transportation system can collect real-time information from GPS devices, traffic sensors, vehicles, weather systems, and road cameras. Streaming technologies can analyze this data continuously to identify traffic congestion and calculate optimal routes. This helps reduce travel time, fuel consumption, delivery delays, and operational costs.

Example: A logistics company can detect a traffic jam on a delivery route and automatically suggest an alternative route to the driver.

10. Ethical and Legal Challenges of Big Data

Big Data creates ethical and legal concerns related to privacy, unauthorized data collection, security breaches, discrimination, and misuse of personal information. Organizations should obtain user consent, implement encryption and access controls, minimize unnecessary data collection, and comply with applicable data-protection regulations. Transparent data policies and regular audits improve responsible data usage and user trust.

Example: A company collecting customer location data should clearly inform users about its purpose and obtain appropriate consent before using it for analytics.